

Supplementary Information: Ephaptic A β -to-C-Fiber Crosstalk and Multi-Fiber Spatial Summation

Mina Saied Attia Rizk

S1 Numerical verification

n	lagged, $\Delta t = 2.5 \mu s$	lagged, $\Delta t = 0.125 \mu s$	monolithic, $\Delta t = 5 \mu s$	monolithic, $\Delta t = 0.125 \mu s$
1	1.39	1.51	1.54	1.51
10	5.01	10.00	10.03	9.94
25	6.10	15.41	15.39	15.47
50	6.28	20.88	21.22	21.72

Table S1. The lagged coupling scheme of the earlier version vs the monolithic scheme, in the earlier full-length geometry and parameters (HH $E_{Na} = +35.64$ mV, 100-nA stimulus inside the compartment, $\Delta z = 10 \mu m$). Entries are the peak C-fiber depolarization between $z = 3$ and 8 mm (mV). The lagged error grows with n ; the monolithic values are Δt -independent to within 2.4 %.

model	n	Δz 20 μm	Δz 10 μm	production (5 μm , 1 μs)	Δz 1.25 μm	Δt 0.125 μs	conductance-implicit
HH	1	1.43	1.53	1.56	1.57	1.55	1.54
HH	10	10.25	10.79	10.93	10.97	10.93	10.89
HH	25	18.88	19.12	19.34	19.40	19.39	19.41
HH	50	22.35	23.26	23.51	23.59	23.62	23.76
NavC	1	1.43	1.53	1.56	1.57	1.55	1.54
NavC	10	10.25	10.79	10.93	10.97	10.93	10.89
NavC	25	18.90	19.13	19.35	19.41	19.40	19.42
NavC	50	22.36	23.28	23.53	23.61	23.63	23.78

Table S2. Grid and time-step refinement of the peak C-fiber depolarization in the lesion (mV) in the production configuration. Unless stated, $\Delta z = 5 \mu m$, $\Delta t = 1 \mu s$, explicit ionic currents. “Conductance-implicit”: ionic conductances in the system matrix.

S2 Membrane verification

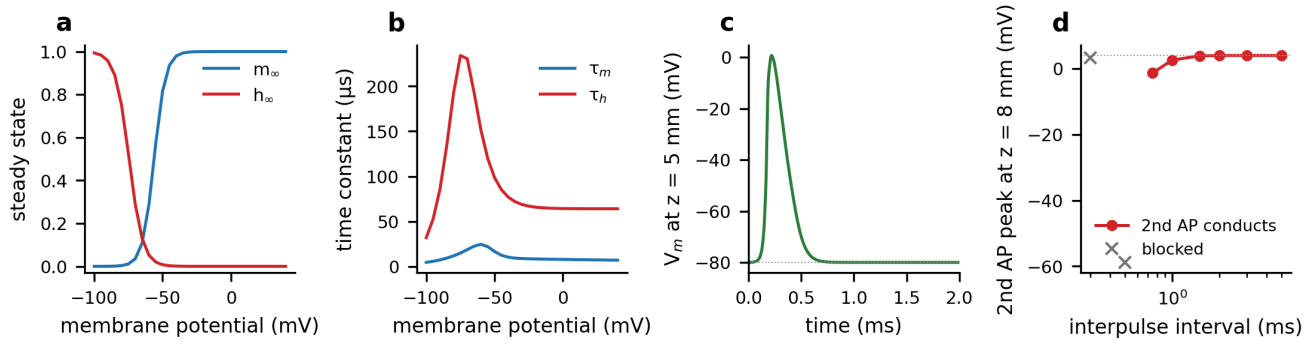


Fig. S3 Verification of the A β source. (a) CRRSS steady-state activation and inactivation and (b) their time constants over the modelled range, at 37 °C. (c) Action potential at a node ($z = 5$ mm) followed through repolarization to rest. (d) Paired-pulse recovery: peak of the response at $z = 8$ mm after a second stimulus, against the interpulse interval. Grey crosses mark intervals at which no second action potential propagates; the value plotted there is the decaying tail of the first action potential, not a second spike.

interval (ms)	APs at $z = 8$ mm	2nd AP peak (mV)	2nd AP CV (m/s)	2nd AP conducts
0.30	1	+3.4	–	False
0.50	1	-58.9	–	False
0.75	2	-1.3	36.0	True
1.00	2	+2.6	39.2	True
1.50	2	+3.9	40.4	True
2.00	2	+4.0	40.4	True
3.00	2	+4.0	40.0	True
5.00	2	+4.0	40.0	True

Table S7. Paired-pulse recovery of the isolated A β fiber ($2 \times$ threshold, 0.2-ms pulses at node 0; counts and peaks read at $z = 8$ mm). The first action potential conducts at 40.0 m/s in every run.

C-fiber membrane	rest (mV)	τ_m at rest (ms)	λ , small signal (μm)	R_{in} predicted ($\text{M}\Omega$)	R_{in} measured ($\text{M}\Omega$)	rheobase, 1 ms (nA)
HH	-65.00	1.48	198	68.9	68.8	0.16
NavC	-66.82	1.37	176	61.5	61.3	0.70

Table S8. Passive properties of the isolated C-fiber membranes under the production geometry. λ and R_{in} are small-signal values, computed from the slope conductance dI_{ion}/dV at rest; R_{in} measured is the steady response to a 1-pA injection at $z = 5$ mm. Rheobase is for a 1-ms point injection at the same site.

S3 Strength–duration characterization

C-fiber membrane	Q _{th} 0.1 ms (pC)	V _{peak} 0.1 ms (mV)	V _{peak} 1 ms (mV)	V _{peak} 20 ms (mV)	I _{th} 0.1 / I _{th} 20 ms
HH, 6.3 °C	0.14	-30	-51	-53	21.3
HH, 25 °C	0.10	-38	-43	-38	5.4
HH, conductances × 5	0.11	-38	-54	-56	12.7
HH, rates & conductances × 5	0.07	-48	-52	-53	4.6
HH, rates & conductances × 10	0.05	-51	-53	-53	2.5
NavC, $\tau_{m8} = 1.5$ ms	0.41	39	-19	-31	7.3
NavC, $\tau_{m8} = 0.05$ ms	0.16	-24	-34	-39	4.0
NavC, time constants ÷ 10, conductances × 10	0.22	-19	-31	-31	1.2

Table S3. Point current injection at $z = 5$ mm into the isolated C-fiber. Q_{th} : threshold charge for a 0.1-ms pulse; V_{peak} : largest membrane potential at the injection site for a pulse 0.5 % below threshold; last column: ratio of threshold currents for 0.1-ms and 20-ms pulses.

S4 Sensitizing bias

model	bias (A/m ²)	rest (mV)	spontaneous AP	peak V, n = 25 (mV)	peak V, n = 100 (mV)	evoked AP (any n)
HH	0	-65.00	no	-45.7	-38.1	no
HH	0.01	-64.20	no	-44.9	-37.3	no
HH	0.02	-63.48	no	-44.2	-36.6	no
HH	0.03	-62.85	no	-43.5	-35.9	no
HH	0.04	-62.27	no	-43.0	-35.4	no
HH	0.05	-61.73	no	-42.4	-34.8	no
HH	0.06	-61.24	no	-41.9	-34.4	no
HH	0.07	-60.78	no	-41.5	-33.9	no
NavC	0	-66.82	no	-47.5	-39.9	no
NavC	0.25	-59.30	no	-40.0	-32.3	no
NavC	0.5	-56.11	no	-36.9	-29.1	no
NavC	1	-52.20	no	-33.1	-25.2	no
NavC	1.5	-49.52	no	-30.5	-22.6	no
NavC	2	-47.39	no	-28.4	-20.6	no
NavC	2.5	-45.55	no	-26.7	-18.8	no
NavC	3	-43.90	no	-25.1	-17.3	no
NavC	3.5	-42.35	no	-23.7	-15.9	no

Table S4. Uniform depolarizing bias current on the C-fiber (production configuration). “Spontaneous AP”: no-stimulus control of 200 ms. Peak V: largest C-fiber membrane potential in the lesion after the A β volley. The HH resting state is linearly stable up to 0.095 A m⁻² and unstable from 0.1 A m⁻²; the NavC resting state stays stable over the whole tested range (to 4 A m⁻²) and a second stable, depolarized state appears above 2.75 A m⁻² (space-clamped analysis with linear stability from the Jacobian, e10_equilibria.csv).

S5 Temporal dispersion: convergence in the number of onset phases

study	K	Δz (μm)	spacing $W/(K-1)$ (μs)	peak ΔV (mV)
phases K	11	20	150	2.23
phases K	21	20	75	1.31
phases K	41	20	38	0.97
phases K	61	20	25	0.91
phases K	81	20	19	0.89
grid Δz	41	20	38	0.97
grid Δz	41	10	38	1.02
grid Δz	41	5	38	1.03

Table S5. Dispersed volley ($n = 25$, $W = 1.5$ ms, NavC). The peak depends on the number of onset phases K until the spacing $W/(K - 1)$ is well below the ~ 0.1 -ms duration of one fiber's contribution; the production runs use a spacing of at most 25 μs . Grid refinement at $K = 41$ changes the peak by 6.2 %, so the sweep is run at $\Delta z = 20$ μm . One classification does depend on that choice: the synchronous $n = 25$ volley is close to A β conduction block, and on $\Delta z = 20$ μm its action potentials still cross the lesion, slowly (13 m/s), where every grid from $\Delta z = 10$ μm down blocks them (Table S2). The peak depolarization differs by only 2.3 % between the two grids, and the attenuations quoted in the text are ratios taken within one grid, so neither the attenuation nor the primary endpoint is affected.

S6 The earlier full-length geometry

electrode current	model	A β stimulus (nA)	propagating C-fiber AP for $n \geq$	initiation (mm)	A β node 0 peak, $n = 25$ (mV)	max
omitted	HH	3.02	none ($n \leq 50$)	–	+17	71
omitted	HH	10	25	0.00	+110	102
omitted	HH	30	10	0.00	+413	276
omitted	HH	100	2	0.00	+1465	935
omitted	NavC	3.02	none ($n \leq 50$)	–	+17	71
omitted	NavC	10	none ($n \leq 50$)	–	+110	102
omitted	NavC	30	none ($n \leq 50$)	–	+414	277
omitted	NavC	100	none ($n \leq 50$)	–	+1466	938
returned	HH	3.02	none ($n \leq 50$)	–	+4	34
returned	HH	10	none ($n \leq 50$)	–	+24	65
returned	HH	30	none ($n \leq 50$)	–	+139	194
returned	HH	100	25	0.07	+556	676
returned	NavC	3.02	none ($n \leq 50$)	–	+4	34
returned	NavC	10	none ($n \leq 50$)	–	+24	65
returned	NavC	30	none ($n \leq 50$)	–	+139	194
returned	NavC	100	50	0.07	+556	676

Table S6. Shared compartment along the whole cable, sealed ends, A β electrode at node 0 inside the compartment ($\kappa = 10^9$ m $^{-2}$, $A_e = 16.4$ μm^2). C-fiber action potentials are launched at the electrode site only with suprathreshold electrode currents; the electrode-site membrane potentials show the non-physiological state produced by 100 nA.

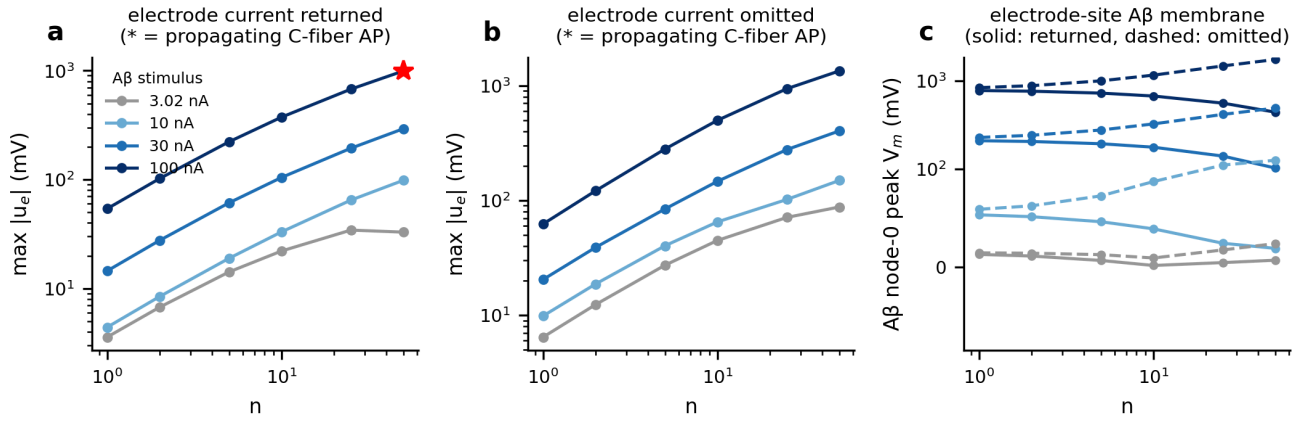


Fig. S1 Earlier full-length geometry. (a, b) Peak C-fiber membrane potential at the electrode site $z = 0$ vs n for A β stimuli of 2 $\times$ threshold to 100 nA (stars: propagating C-fiber AP launched at the electrode). (c) Peak A β membrane potential at the stimulated node.

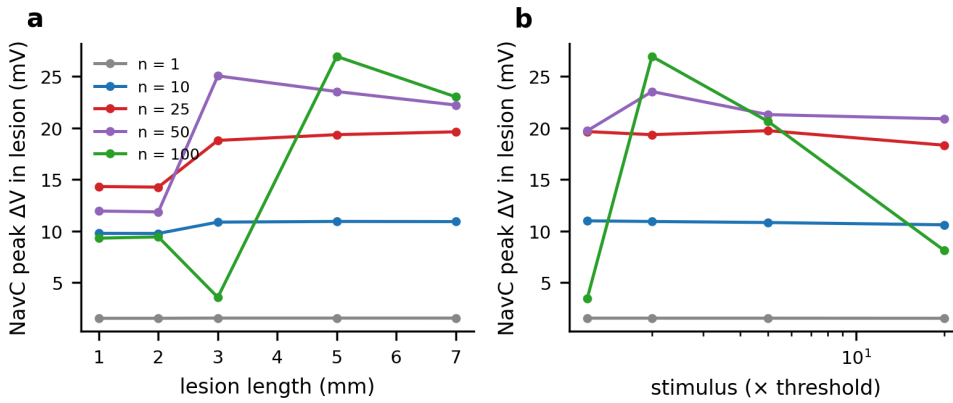


Fig. S2 Lesion length (a) and A β stimulus strength (b) in the production configuration (NavC). No C-fiber action potential occurred.